

**FIFTH SEMESTER U.G. (CUCBCSS—UG) DEGREE [SPECIAL]
EXAMINATION, NOVEMBER 2020**

Mathematics

MAT 5B 06—ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 06—ABSTRACT ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. Which of the following defines a binary operation on $\mathbb{Z}^+$?
 - (A) $a * b = a - b$.
 - (B) $a * b = c$, where c is the smallest integer greater than both a and b .
 - (C) $a * b = c$, where c is at least 5 more than $a + b$.
 - (D) $a * b = c$, where c is the largest integer less than the product of a and b .
2. If b and c are the inverses of some element a in a group G then :
 - (A) $b = c$.
 - (B) $b \neq c$.
 - (C) $b = kc$ for some $k \in \mathbb{N}$.
 - (D) None of these.
3. On $\mathbb{Q}$, which of the following does not define a binary operation ?
 - (A) $a * b = |a| |b|$.
 - (B) $a * b = (a - b)^2$.
 - (C) $a * b = +\sqrt{ab}$.
 - (D) None of these.
4. Let $*$ be the binary operation defined on $\mathbb{Q}^+$ as $a * b = ab/2$. Then inverse of the element a is :
 - (A) $2a$.
 - (B) $4/a$.
 - (C) a^2 .
 - (D) None of these.
5. Which of the following are true ?
 - (1) A group may have more than one identity element.
 - (2) Any two groups of three elements are isomorphic.
 - (3) Every group of at most three elements is abelian.
 - (A) 2 and 3.
 - (B) 1 and 2.
 - (C) 1 and 3.
 - (D) All.
6. Let $G = \{1, -1, i, -i\}$ where, be a set of four elements. Which of the following is a binary operation on G ? (1) $a * b = a + b$. (2) $a * b = a.b$.
 - (A) Only 1.
 - (B) Only 2.
 - (C) Both.
 - (D) None of these.

7. In a group G , $(a * b)^2 = a^2 * b^2$ for all $a, b \in G$. This statement is :

- (A) Always true.
- (B) True if G is finite.
- (C) True if G is a multiplicative group.
- (D) True if G is abelian.

8. If a group G is of order 31, then which of the following is false ?

- (A) G is abelian.
- (B) G is cyclic.
- (C) G is abelian but not cyclic.
- (D) Both abelian and cyclic.

9. The Klein 4-group is isomorphic to _____.

- (A) $Z_2 \times Z_4$.
- (B) $Z_2 \times Z_2$.
- (C) Z_4 .
- (D) None of these.

10. Order of $(2, 2)$ in $Z_4 \times Z_6$ is _____.

- (A) 2
- (B) 6.
- (C) 4.
- (D) 12.

11. Which of the following is true ?

- (A) Every cyclic group has a unique generator.
- (B) In a cyclic group, every element is a generator.
- (C) Every cyclic group has at least two generators.
- (D) None of these.

12. Which of the following is a cyclic group with only one generator ?

- (A) Z_2 .
- (B) $(Z, +)$.
- (C) Klein-4 group.
- (D) None of these.

13. Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$. Then σ^6 equals :

- (A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$
- (B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix}$
- (C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 4 & 5 & 6 & 2 \end{pmatrix}$
- (D) None of these.

14. The product $(1\ 3\ 6)(2\ 4)$ of two permutation is :

- (A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$. (B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 2 & 5 & 4 & 6 \end{pmatrix}$.
- (C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 6 & 1 & 4 \end{pmatrix}$. (D) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 4 & 6 & 1 \end{pmatrix}$.

15. Which of the following is true ?

- (A) Every function is a permutation if and only if it is one to one.
 (B) The symmetric group S_3 is cyclic.
 (C) The symmetric group S_n is not cyclic for any n .
 (D) Every function from a finite set onto itself must be one to one.

16. Which of the following is an even permutation ?

- (A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 1 & 6 & 2 & 1 & 8 \end{pmatrix}$. (B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 4 & 5 & 3 & 7 & 8 & 6 \end{pmatrix}$.
- (C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 4 & 3 & 5 & 2 & 6 & 8 & 7 \end{pmatrix}$. (D) None of these.

17. What is the largest possible order of a cyclic subgroup of $Z_{12} \times Z_{15}$?

- (A) 60. (B) 30.
 (C) 180. (D) None of these.

18. In a non-abelian group the element a has order 108. Then the order of a^{12} is :

- (A) 54. (B) 27.
 (C) 18. (D) 9.

19. f is a homomorphism $f: (R, +) \rightarrow (Z, x)$ such that $f(2) = 3$. Then $f(6)$ is :

- (A) 6. (B) 9.
 (C) 18. (D) 27.

20. Which of the following is not true ?

- (A) Every subgroup of every group has left cosets.
 (B) A subgroup of a group is a left coset of itself.
 (C) A_n is of index 2 in S_n for $n > 1$.
 (D) None of these.